

THE HAZARD RATE & MEMORYLESS PROPERTY OF EXPONENTIAL DISTRIBUTION

Given Exponential Distribution

$$P(T=t) = \begin{cases} \lambda e^{-\lambda t} & t \geq 0 \\ 0 & t < 0 \end{cases}$$

MEMORYLESS PROPERTY

$$P(T > t+s | T > s) = P(T > t)$$

Issue: Markets have memory!

HAZARD RATE

Measures instantaneous prob. happening now, given it has not happened yet.

$$\begin{aligned} P(T \leq t+dt | T > t) &= 1 - P(T > t+dt | T > t) = \\ &= 1 - P(T > dt) = 1 - e^{-\lambda dt} = 1 - \underbrace{\left[1 - \lambda dt + \frac{1}{2} \lambda^2 dt^2 - \dots \right]}_{\approx \lambda dt} \end{aligned}$$

$$P(t < T \leq t+dt) = \lambda P(T > t) dt$$

Issue: If we model default risk, we assume default risk in year 30 same for day 1.